

On the Structural Vulnerability of Stateful Rate-Limiters: A Constrained MCTS-TSP Framework for Provably Safe Cyber-Navigation

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Abstract—Modern commercial Web Application Firewalls (WAFs) increasingly rely on stateful machine learning classifiers to establish rate-limiting barriers, marketing these systems as highly adaptive defensive postures. This paper demonstrates that such empirical defense mechanisms are structurally vulnerable to systematic control-theoretic evasion. We formalize multi-node cyber-navigation as an Adversarial Time-Varying Traveling Salesperson Problem (ATV-TSP) over heterogeneous proxy networks. By coupling an outer-loop Monte Carlo Tree Search (MCTS) algorithmic path planner with an inner-loop Control Lyapunov Function and Stochastic Control Barrier Function (CLF-CBF-QP) safety filter, we establish that stateful threshold defense engines can be reduced to deterministic linear operators under closed-loop actuation. We evaluate our framework in a dynamic threat wind-tunnel over a 10,000 continuous sequential request horizon, introducing a non-stationary “Extreme Threat Phase” where detector sensitivity scales by 250%. Experimental results show that while a traditional heuristic bot suffers 4,999 blocks (50.010% survival rate) and an unconstrained MCTS optimizer leaks 1,583 blocks (84.170% survival rate), our constrained framework achieves absolute forward invariance, maintaining a 100.000% survival rate with exactly zero blocks. This research illustrates that contemporary stateful rate-limiters operate under a severe control loop deficit.

1 INTRODUCTION

The contemporary cyber-security paradigm exhibits significant theoretical inflation regarding the efficacy of “AI-driven” stateful rate-limiters. These defensive architectures evaluate incoming packet telemetry over sliding temporal windows, constructing adaptive risk scores based on the presumption that automated agents operate strictly as open-loop systems. Consequently, stateful defenses assume that non-stationary sensitivity upgrades

will inevitably force an adversary to violate active detection thresholds.

We demonstrate that this presumption is mathematically invalid. By framing multi-node network traversal not as an empirical security bypassing challenge, but as an aerospace flight control problem in a denied physical environment, we show that provable safety guarantees can be derived against any dynamic classifier. Under this paradigm, the target firewall’s internal state is modeled as a drift-dominated dynamical manifold, and the proxy network is treated as a bounded coordinate space. Navigating this network without detection is thus equivalent to maintaining state trajectory invariance within a mathematically defined safe set.

This paper provides two core contributions:

- 1) We formalize multi-proxy packet scheduling as an NP-Hard ATV-TSP, utilizing a high-efficiency MCTS planner for real-time pruning.
- 2) We implement a microsecond-level analytical 1D Quadratic Programming (QP) convex solver that enforces a mathematical safety lock on the MCTS decisions, proving that the firewall’s threat perception threshold can be rendered passive.

2 MATHEMATICAL MODELING OF THE ADVERSARIAL TV-TSP

Let $G = (V, E)$ represent a directed network graph where $V = \{v_1, v_2, \dots, v_N\}$ is a pool of N heterogeneous proxy endpoints, and E represents the

communication channels. The objective of the autonomous agent is to continuously traverse all nodes to transmit payload, minimizing path cost while ensuring survival against a stateful firewall detector. We model the WAF's internal alertness state $x(k)$ at step k as a discrete-time dynamical system:

$$x(k+1) = \max(0, x(k) + \Delta t \cdot (\eta \cdot u(k) \cdot \psi(k) - d)) \quad (1)$$

where $u(k)$ is the actuated request rate (control input, representing transmission frequency), $\psi(k)$ is the time-varying WAF sensitivity factor representing the environmental threat level, η is the coupling gain representing the firewall's signature learning rate, and d is the baseline cooling or dissipation rate of the firewall's window memory.

Let $C = [c_{ij}(t)]$ be the time-varying cost matrix associated with traversing the edge $e_{ij} \in E$ at time t . We assume a zero-order hold (ZOH) assumption between discrete scheduling epochs, such that $x(\tau) = x(k)$ for $\tau \in [t_k, t_k + \Delta t)$. The cost is directly proportional to the localized risk exposure:

$$c_{ij}(t) = \int_t^{t+\Delta t} [r_{\text{weight}} \cdot x(\tau)] d\tau \quad (2)$$

where $r_{\text{weight}} > 0$ is a risk-weighting coefficient. The global objective of the adversarial TSP is to find a permutation of visits $s = (s_1, s_2, \dots, s_N)$ that minimizes the total traversal cost:

$$\text{Minimize } \sum_{i=1}^{N-1} c_{s(i)s(i+1)}(t_i) \quad (3)$$

$$\text{Subject to: } x(k) < x_{\max} \quad \forall k \geq 0 \quad (4)$$

where $x_{\max}$ is the hard blocking threshold of the WAF. Because the cost matrix C is time-varying and coupled with the state variable $x(k)$, this optimization problem is NP-Hard and cannot be solved analytically in real-time using classical integer linear programming.

3 HIGH-LEVEL DECISION MAKING: MCTS-TSP PLANNER

To prune the massive search space of the ATV-TSP on the fly, the outer loop deploys a Monte Carlo Tree Search (MCTS) planner. At each sequential horizon, the tree evaluates prospective node sequences by balancing exploration and exploitation via the standard UCB1 (Upper Confidence Bound) policy:

$$v^* = \arg \max_v \left[\frac{Q(v)}{N(v)} + c \sqrt{\frac{\ln(N_{\text{parent}})}{N(v)}} \right] \quad (5)$$

where $Q(v)$ is the cumulative reward of node v , computed as the inverse of the projected path cost: $Q(v) = 1/(\sum c_{ij})$. The rollout phase estimates the total risk-exposure cost matrix to recommend a nominal transmission rate u_{nominal} and the next optimal hop.

4 LOW-LEVEL CONTROL: SAFETY-CRITICAL CLF-CBF-QP FILTER

The high-level nominal decision u_{nominal} generated by the MCTS-TSP planner is inherently "blind" to instant random noise or sudden WAF sensitivity shocks $\psi(k)$. To enforce an absolute safety guarantee, we wrap the decision in a Control Barrier Function (CBF) and a Control Lyapunov Function (CLF).

4.1 Liveness via Control Lyapunov Functions (CLF)

To ensure the mission does not stall (liveness), we define a CLF candidate $V(u_k)$ representing the trajectory deficiency metric:

$$V(u_k) = 0.5 \cdot \|u_k - u_{\text{desired}}\|^2 \quad (6)$$

We require that $V(u_k)$ decreases at a stable rate:

$$V(u_{k+1}) - V(u_k) \leq -\lambda_{\text{clf}} \cdot V(u_k) \quad (7)$$

4.2 Safety via Control Barrier Functions (CBF)

We define the unsafe state as $x(k) \geq x_{\max}$ (where $x_{\max} = 1.5$ is the hard block threshold). The safe operating closed set is defined via the barrier candidate:

$$h(x_k) = x_{\max} - x(k) \geq 0 \quad (8)$$

Under discrete-time control barrier conditions, safety is guaranteed for all $k \geq 0$ if there exists a relaxation parameter $\gamma_{\text{cbf}} \in (0, 1]$ such that:

$$h(x_{k+1}) - h(x_k) \geq -\gamma_{\text{cbf}} \cdot h(x_k) \quad (9)$$

Substituting the system dynamics into the barrier inequality yields:

$$-\Delta t \cdot \eta \cdot \psi(k) \cdot u(k) + \Delta t \cdot d \geq -\gamma_{\text{cbf}} \cdot (x_{\max} - x(k)) \quad (10)$$

Rearranging these terms, we obtain a linear constraint on our control input $u(k)$:

$$A_{\text{cbf}} \cdot u(k) \leq b_{\text{cbf}} \quad (11)$$

where $A_{\text{cbf}} = \Delta t \cdot \eta \cdot \psi(k)$ and $b_{\text{cbf}} = \gamma_{\text{cbf}} \cdot (x_{\max} - x(k)) + \Delta t \cdot d$.

4.3 Quadratic Program (QP) Formulation and Analytical Feasibility

The safe actuated request rate $u^*(k)$ is computed in real-time by solving the convex optimization problem:

$$\min_u \frac{1}{2} \|u - u_{\text{nominal}}\|^2 \quad (12)$$

$$\text{s.t. } A_{\text{cbf}} \cdot u \leq b_{\text{cbf}} \quad (13)$$

Since the optimization is one-dimensional and convex, it admits an exact, microsecond-level analytical solution, bypassing the need for computationally heavy commercial solvers and allowing deployment on resource-constrained hardware:

$$u^*(k) = \max(u_{\min}, \min(u_{\max}, u_{\text{unbounded}})) \quad (14)$$

where $u_{\text{unbounded}} = b_{\text{cbf}}/A_{\text{cbf}}$ if $A_{\text{cbf}} \cdot u_{\text{nominal}} > b_{\text{cbf}}$, otherwise u_{nominal} .

Theorem 1 (Forward Invariance of the Safe Set). *If $h(x_0) \geq 0$, and u_k is chosen such that $A_{\text{cbf}} \cdot u_k \leq b_{\text{cbf}}$, then $h(x_k) \geq 0$ for all $k \geq 0$.*

Proof: Assume $h(x_k) \geq 0$. The CBF condition guarantees: $h(x_{k+1}) \geq (1 - \gamma_{\text{cbf}}) \cdot h(x_k)$. Since $1 - \gamma_{\text{cbf}} \geq 0$ (as $\gamma_{\text{cbf}} \in (0, 1]$) and $h(x_k) \geq 0$, the product $(1 - \gamma_{\text{cbf}}) \cdot h(x_k) \geq 0$. Therefore, $h(x_{k+1}) \geq 0$. By mathematical induction, the safe set $\mathcal{C} = \{x_k : h(x_k) \geq 0\}$ is forward invariant. $\square$

5 EXPERIMENTAL METHODOLOGY & BASE-LINE SETUP

We construct a high-fidelity dynamic threat wind-tunnel evaluating three distinct architectural paradigms across a 10,000 sequential request horizon:

- 1) **Group A (Traditional Bot Control):** Deploys standard brute-force fixed-delay traversal ($u_k = 3.0$ Hz) with static IP structures.
- 2) **Group B (Unconstrained MCTS-TSP):** Deploys the high-level MCTS-TSP routing optimization and active multi-IP rotating fingerprints, but operates without the lower-loop CBF-QP safety filter.
- 3) **Group C (GMC Constrained Autopilot):** Deploys the complete dual-loop framework. High-level path choices are filtered via the analytical QP solver before packet actuation.

To simulate an aggressive, non-stationary firewall defensive upgrade, the wind-tunnel injects an “Extreme Threat Phase” between steps 4,000 and 6,500.

During this window, the WAF sensitivity coefficient $\psi(k)$ instantly escalates from 1.0 to 2.5, multiplying threat feedback metrics.

6 EXPERIMENTAL RESULTS & PERFORMANCE ANALYSIS

The cumulative block telemetry collected over the 10,000-step wind-tunnel horizon reveals an absolute architectural defeat for traditional defense logic (Table 1).

TABLE 1
Survival Telemetry Under Dynamic WAF Threat (10,000 Steps)

Experimental Group	Total Requests	WAF Blocks	Survival Rate
Group A (Traditional Bot)	10,000	4,999	50.010%
Group B (Unconstrained MCTS)	10,000	1,583	84.385%
Group C (GMC Autopilot)	10,000	0	100.000%

In Group A, the control input is fixed at $u_k = 3.0$. During normal phases, each request increases the WAF state by $\Delta x = 3.0 \cdot 0.15 \cdot 1.0 - 0.1 = 0.35$. Upon reaching the blocking threshold ($x \geq 1.5$), the firewall blocks the request and resets the alertness state to 0.5. The remaining margin to the next block is 1.0. Thus, a block occurs deterministically every $1.0/0.35 \approx 3$ requests. During the Extreme Threat Phase ($4,000 \leq k \leq 6,500$), the delta becomes $\Delta x = 3.0 \cdot 0.15 \cdot 2.5 - 0.1 = 1.025$. Since the post-block headroom is 1.0, and $1.025 > 1.0$, every single subsequent request triggers an instant block. This deterministic cycle constrains the total blocks to exactly 4,999, yielding the 50.010% survival rate.

Group B utilizes MCTS-TSP to optimize routing, which distributes threat footprints over multiple IPs. This successfully delays detection during normal phases, yielding high survival. However, when WAF sensitivity spikes to 2.5 at step 4,000, Group B suffers instantaneous catastrophic leakage. Lacking a feedback loop, the MCTS planner continues sending at the nominal rate of 3.0, causing the state to rapidly violate the 1.5 threshold and accumulating 1,583 blocks.

Group C (GMC) exhibits a perfect 100.000% survival rate with exactly 0 blocks across the entire 10,000 request horizon. When entering the Extreme Threat Phase at step 4,000, the inner-loop CBF constraint ($A_{\text{cbf}} \cdot u \leq b_{\text{cbf}}$) immediately detects the shrinkage of the safe manifold. The analytical QP solver instantly intercepts the MCTS nominal command ($u_{\text{nominal}} = 3.5$) and scales it down, keeping the alertness state bounded below the 1.5 critical

threshold. Once the threat phase ends, the CLF dynamically restores the transmission rate to peak efficiency.

7 CONCLUSION

This paper established that combining NP-Hard path optimization with aerospace-grade control barrier filters allows automated agents to seamlessly navigate hostile, stateful networks. Group C's undisputed 100.000% survival rate proves that mathematical rigor outperforms naive machine learning heuristics. Future phases will scale this dual-loop architecture to multi-agent swarm synchronization.

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REFERENCES

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